

1. Description

2/C FLEXO is a two pot alcohol based ink specifically designed for surface printing. The ink can be printed flexo and gravure

2. Product features

2/C FLEXO inks:

- Have excellent printability
- Specifically designed for surface printing
- Have immediate excellent adhesion onto several substrates
- Are chlorine free
- Have to be printed within 6 hours from the blending with the hardener to maintain the properties. In case of re-use of the same ink, the day after, it is recommended to add again the hardener, no more than one time. An excess of hardener gives a printed film too sticky.
- Have a pot life in the diluted form of 6-7 days before hardening into the pot
- Prints are suitable to the pasteurization process. The boiling resistance is achieved by overprinting colours with the two pot lacquer cod. 9W8200203. It is mandatory to process thermically any print after 7 days to complete the cross-linking of the ink. It is important to print films which guarantee a functional barrier between food and substrate when a hot process is requested. Due to the different substrates, foods and hot processes , a preliminary test by the customer is suggested to confirm all the specifications requested
- We suggest to wash and clean cylinders and anyloxes at any stop of the press to avoid the irreversible occlusion of cells
- No compatible with other ink series
- A careful cleaning of the press is needed when inks series is damped.

MIXING RATIO:

INK	cod. 9-W52:	100
HARDENER	cod. 9-W910-3200:	10



3. Product Suitability

3.1. Applications

2/C FLEXO inks can be used for surface prints with good general resistances on several substrates

The high content of ethanol of this series and the long pot life at the press is the key factor to allow the flexo- printing with a two-pot system

7 days after printing , the cross linking of the print is complete and an high heat resistance and flexibility is guaranted in case of heat sealing at 200/220 °C

3.2. Substrate

2C/FLEXO inks are suitable for the following substrates:

- PE Corona treated > 38 dynes/cm
- OPP Coex – Plain corona treated > 38 dynes/cm
- PET Corona treated > 48 dynes/cm
- PA (Nylon) Corona treated > 48 dynes/cm
- PA Chemically treated used as skin tubes for sausages (we always suggest a preliminary test or an evaluation by our lab team)

4. Colour Range

2C/FLEXO inks are available as finished inks in the following C.I.:

P.W.6	WHITE	Base 100
P.Y.13	Medium Yellow	Base 236
P.Y.83	Warm Yellow	Base 234
P.O.34	Orange	Base 330
P.R.166	Warm Red	Base 408
P.R.184	Rubine	Base 442
P.B.15:4	Cyan	Blue Base 600
P.G.7	Green C	Base 720
P.V.37	Violet	Base 572
P.BL.7	Black	Base 800

Extender Varnish 9W8200203

HARDENER 9W9103200



5. General Handling

5.1. Storage

- 2/C flexo inks are considered highly flammable. Suitable precautions should be observed to avoid ignition sources.
- Temperature of storage: from 5 °C TO 35 °C
- Shelf life: 12 months under above conditions

5.2. Waste disposal

Care should be exercised in the disposal of printing ink waste. This should be carried about in accordance with good industrial practice, observing all the appropriate regulations and guidelines.

For more specific handling advice refer to the Safety Data Sheet (MSDS)

6. Printing Conditions

6.1. Printing Viscosities

Inks must thoroughly be stirred and homogeneous before reducing with solvent. The optimum printing viscosity depends on press and running conditions. Good results have been achieved with 2/C FLEXO inks at the following viscosities:

- Rotogravure : 13-15 sec (Din cup N°4)
- Flexo : 16-25 sec (Din cup N°4)]

6.2. Reducing Solvents

- **Flexo :**

Normal: Ethanol / Ethylacetate blends 80/20
Retarder : n-Propanol / Methoxy Propanol

- **Rotogravure :**

Rapid: Ethylacetate / Ethanol blends 1/1
Normal : Ethanol
Slow : n-Propanol



6.3. Wash up solvents

Mixture Alcohols / Esters

6.4. Cylinders / Plates

2/C FLEXO inks are suitable for all types of anilox and cylinders (chemical and mechanical engraving) and different types of plates.

7. End-use safety

Acceptable technical performance of 2/C FLEXO inks is dependent on:

- Control of film weight (inks and adhesives)
- Adequate drying on press (to ensure that the print is dry before conversion)
- Control of any migratable substances within the print, film or adhesives

Choice and control of film weight, drying, substrate and adhesives are printer technical requirements for which the Sun Chemical technical team can provide assistance in the form of suggestions or direct support.

Food Packaging:

It is the ultimate responsibility of the food packaging supply chain to ensure packaged food is safe. With regard to potential migration of substances in to food from the printed packaging, it is the responsibility of the converter and food packager to ensure that no such migration occurs beyond permitted regulatory levels.

To fulfill its responsibility within the supply chain Sun Chemical will provide details to the customer of potential migratable components, where present in its inks, through the preparation and communication of a 'Statement of Composition'

For the **2/C FLEXO ink** series:

- There are no intentionally added migratable components within the ink formulation
- Or, there are specific additives with potential to migrate within the ink formulation – details of these are available to customers through copies of Statement of Composition



8. Disclaimers

The ink 2/C FLEXO is not intended to direct food contact.

This list of applications, substrates and processes provided in this document is not exhaustive. Please contact your local Sun Chemical representative for full technical evaluation of your application or process.

9. Technical Assistance / Contacts

For further information, please contact your local Sun Chemical team.

Our Products are intended for sale to professional users. The information herein is general information designed to assist customers in determining the suitability of our products for their applications. All recommendations are made without guarantee, since the application and conditions of use are beyond our control. We recommend that customers satisfy themselves that each product meets their requirements in all respects before commencing a print run. There is no implied warranty of merchantability or fitness for purpose of the product or products described herein. In no event shall Sun Chemical be liable for damages of any nature arising out of the use or reliance upon this information. Modifications of the product for reasons of improvements might be made without further notice.

August 2014/ Rev.2

